just sell seven cryogenic engines, and not transfer the technology. That this delayed India's cryogenic programme by at least fifteen years is another tale.

So, to make HS-25, I had nowhere to go, but MIDHANI, Hyderabad. I understood that Abdul Kalam, who had by now developed SLV-3, had asked MIDHANI to deliver another material called maraging steel. This alloy, which can withstand high temperature and pressure, also has some characteristics that qualify it to be used in the first stage of PSLV boosters.

Already struggling to deliver maraging steel, MIDHANI said it would be impossible to develop the equivalent of HS-25. But I wouldn't give up. After several rounds of cajoling, I used my last weapon: threat. I conveyed it to the MIDHANI brass that India's space programme was a pet project of the prime minister and, as a government body like ISRO, MIDHANI was duty-bound to deliver the material. Any complaint to the PMO would be taken so seriously that the MIDHANI honchos can expect adverse remarks on the service records.

That made the MIDHANI heads consider the proposal, but they still needed a vigorous push. It was here that I became close to my ISRO colleague Sasikumar, who would later be a co-accused with me in the 'ISRO spy case'. Sasi, working with Kalam, was liaising with MIDHANI for maraging steel, and had established some rapport with the MIDHANI men. I requested him to use his connections to push for HS-25 too.

Sasi succeeded in convincing some of them, and put me on to some crucial guys. It wasn't easy, but after several months of pushing and prodding, they agreed to work on it. For MIDHANI it should have been a great realisation of its strengths and capabilities when it delivered the material. The company continues to be the sole supplier of maraging steel for Indian PSLVs and the HS-25 equivalent for ISRO's rockets.

Some other things—like the silica phenolic material used for making the throat of the nozzle—we had to do it with the resources at our disposal at VSSC. The throat, the connecting part between the combustion chamber and the upper part of the nozzle would experience the highest temperature during the firing, and this needed the composite material. REPLACE, a facility of VSSC in Vattiyoorkavu in Trivandrum, was asked to do this. REPLACE was an offshoot of a group working on fibre glass and composites used for making rocket nose cones and fins.